Empirical Volatility Models

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1 GARCH Models

For a review see Chapter 49 of Handbook.

Definition 1. Notation

$$E_{t-1}(\varepsilon_t(\theta_0)) = 0 \tag{1}$$

$$h_t = \sigma_t^2(\theta_0) := E_{t-1}\left(\varepsilon_t^2(\theta_0)\right) \tag{2}$$

$$z_t(\theta_0) = \varepsilon_t(\theta_0)\sigma_t^2(\theta_0^{-1/2}) \tag{3}$$

Usually $\varepsilon_t = y_t - \mu_t$.

Proposition 1.1. *Some Features of Asset Returns*

1. *Thick tails; leptokertic;*
2. *volatility clustering; related to leptokertic;*
3. *leverage effects: δr_t negatively correlated with δv_t.*
4. *information accumulates more slowly when the markets are closed than when they are open.*
5. *forecastable releases of important information are associated with high ex ante volatility.*
6. *inverse relation between volatility and serial correlation for U.S. stock indices.*
7. *Co-movements in volatilities*
8. *stock volatility rises sharply during recessions and financial crises and drops during expansions, the relation between macroeconomic uncertainty and stock volatility is **debatable**. Is there new evidence?*

1.1 Univariate GARCH

1.1.1 ARCH(q)

$$h_t = \sigma_t^2 = \omega + \sum_i^q \alpha_i \varepsilon_{t-i}^2 = \omega + \alpha(L)\varepsilon_{t-1}^2 \tag{4}$$

1.1.2 GARCH(p,q)

$$h_t = \omega + \alpha(L)\varepsilon_{t-1}^2 + \beta(L)h_{t-1}^2 \tag{5}$$

Not suited for leverage effects because of symmetric among signs.

1.1.3 EGARCH

$$\log h_t = \omega + \left(1 + \sum^{q} \alpha_i L^i\right)\left(1 - \sum^{p} \beta_j L^j\right)^{-1} \{\theta z_{t-1} + \gamma[|z_{t-1}| - E|z_{t-1}|]\} \tag{6}$$

1.1.4 GARCH-M

$$\mu_t(\theta) = g[\sigma_t^2(\theta), \theta] \quad \text{in simplest case} \quad g = \mu + \delta\sigma_t^2 \tag{7}$$

1.1.5 Nonparametric GARCH

$$y_t^2 = f(y_{t-1}, y_{t-2}, \ldots, y_{t-p}, \theta) + \eta_t \tag{8}$$

problem: curse of dimensionality; η_t highly heteroscedastic.

See also G-QTARCH(1) model and PNP model(partially nonpara). Also some series estimators(need check out). Also, I wonder if Johnstone's spectral method can be applied, so as to estimate for some main frequency instead of covariates.

1.2 Multivariate Models

$$E_{t-1}(\varepsilon_t) = 0 \quad \text{and} \quad E_{t-1}(\varepsilon_t\varepsilon_t') = \Omega_t \tag{9}$$

Proposition 1.2. *There are some considerations.*

1. *Dimension reduction.*
2. *Does the reduction impose untested restrictions.*
3. *Positive-definite?*
4. *Granger causality.*
5. *Whether the correlations are time-varying.*

6. *If there are portfolios with less persistence.*

7. *Multivariate asymmetric effects?*

1.2.1 Vector ARCH, Diagonal GARCH and BEKK

$$\text{vech}\left(\Omega_t\right) = W + \sum^{q} A_i \,\text{vech}\left(\varepsilon_{t-i}\varepsilon'_{t-i}\right) + \sum^{p} B_j \,\text{vech}(\Omega_{t-j}) \tag{10}$$

Baba(interesting), Engle, Kraft and Kroner, or BEKK model for ensuring positive definiteness.(I wonder if we can do something about things that are not required to be positive and always exists. Is there a CopulARCH?)

$$\text{BEKK see (6.5) in chapter 49} \tag{11}$$

1.2.2 Factor ARCH

First factor model in return, then ARCH-type model for the factor volatility.

$$y_t = B\xi + \varepsilon_t \quad \text{and} \quad \text{var}_{t-1}(\varepsilon_t) = \Psi(\text{time invariant}), \text{var}_{t-1}\left(\xi\right) = \Lambda_t \tag{12}$$

Then

$$\text{var}_{t-1}(y_t) = \Omega_t = \Psi + B\Lambda_t B' \tag{13}$$

Assume Λ is diagonal, we have

$$\Omega_t = \Psi + \sum \beta_k \beta'_k \lambda_{kt} \tag{14}$$

cf. Diebold and Nerlove model:

$$\Omega_t = \Psi + \sum \beta_k \beta'_k \delta^2_{kt} \tag{15}$$

Then there we can construct factor-representing portfolios ϕ_k for each k, such that $\phi'_k \beta_j = \delta_{jk}$.

$$\text{var}_{t-1}(\phi'_k y_t) = \phi'_k \Omega_t \phi_k = \psi_k + \lambda_{kt} \tag{16}$$

Then it's assumed that λ_{kt} has a GARCH(1,1) representation, i.e. there is a set of factor-representing portfolios that follow GARCH(1,1). This is a special case of BEKK

model.

See Orthogonal GARCH in R. Engle, 2002, p.340. Where we find $A'y_t$ such that $V = E(A'y_t y_t' A)$ is diagonal. Fit GARCH for each component of $A'y_t$, get V_t, then

$$H_t = A'^{-1} V_t A^{-1} \tag{17}$$

1.2.3 CCC

Bollerslev 1990. Let D_t be the diagonal of Ω_t, $(\Gamma_t)_{ij}$ be the conditional correlation. Now assume $\Gamma_t = \Gamma$, constant over time.

$$H_t = \Omega_t = D_t^{1/2} \Gamma D_t^{1/2} \tag{18}$$

1.2.4 DCC

From R. F. Engle, Ledoit, and Wolf, 2019. For each i, D_{it} is GARCH(1,1), (here notation is modified to be compatible with previous).

$$D_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i D_{i,t-1} \tag{19}$$

One way to specify the dynamic of Q_t is to assume it follows GARCH(1,1).

$$Q_t = (1 - \alpha - \beta)C + \alpha z_{t-1} z_{t-1}' + \beta Q_{t-1} \tag{20}$$

$$\Gamma_t = \operatorname{diag}(Q_t)^{-1/2} Q_t \operatorname{diag}(Q_t)^{-1/2} \tag{21}$$

1. GARCH for each time series; standardize, get z_t;
2. Estimate unconditional correlation matrix C;
3. Maximize likelihood(easier way is using composite(pairwise) likelihood) and estimate the dynamic.

1.2.5 Shrinkage DCC

It's about a better estimate for C when N is large relative to T. Shrinkage method: recovering population eigenvalues from sample eigenvalues by inverting Marcenko-

Pastur law.

$$\widehat{C} = \frac{1}{T} ZZ' = \sum^{N} \lambda_i u_i u_i' \tag{22}$$

Calculate the population eigenvalues from the sample λ_i by reverting the so-called QuEST function(because for each population (t), we can have a deterministic equivalent of the sample eigenvalues q), so

$$\tau := \arg\min_t \frac{1}{N} \sum^{N} \left[q^i(t) - \lambda_i \right]^2 \tag{23}$$

Then find the shrinked $\tilde{\lambda}$, which will consistently estimate the out-of-sample covariance matrix:

$$\tilde{C} := \sum^{N} \tilde{\lambda}_i u_i u_i' \tag{24}$$

Remark. Simple shrink is the *linear shrinkage*, where $\bar{\lambda}$ is the cross-sectional average of λ, then

$$\tilde{\lambda} = \left[\rho \bar{\lambda} + (1 - \rho) \lambda_i \right] u_i u_i' \tag{25}$$

2 Stochastic Volatility Models

3 Inferences

3.1 Time Series Properties

See X.Chen and Carrasco for mixing and moment properties.

3.2 Estimation

3.3 Testing